

7. GOVERNANCE, LINEAGE & RELEASE GATE

STPA Hazard Mitigation · Bumpless Human Joystick Override · Defensible Release Case (LOOP-01 → REL-01)



1. SENSING

Photons / Voltages to DMA
MIPI CSI-2 · I2C · SPI Bus Priority



2. PERCEPTION

Encoders to Spatial Tokens
ViT · DINOv2 · 3D Affordances



3. WORLD MODELS

Latent Belief & SE(3) Trees
JEPa / RSSM Dynamics · TTL Validity



SYSTEM 1.5 · LINUX MPU / NPU



5. TRAJECTORY DECODERS

Diffusion Policies · ACT Action Chunking (H -steps) · Jerk Bounds



SYSTEM 2 · LINUX MPU



4. SEMANTIC DELIBERATION

VLMs & Spatial Foundation Models · Goals · Leases (t_{expire})



Expiring Proposal p_t



THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY (NO DIRECT MOTOR ACCESS)



SYSTEM 1 · REAL-TIME MCU (BARE-METAL / FreeRTOS)



6. REAL-TIME REFLEX & SAFETY ENFORCEMENT

1 kHz Reflex Timing Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}}(v_t) \leq d_{\text{clearance}}$ · Veto (u_t)



Permitted Action $u_t = \text{permit}(p_t)$



THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$)

Kinetic Momentum ($p=mv$) · Joule Heat Dissipation (I^2R) · Matter Mutation · Friction (μ) · Collision Dynamics

Endogenous Sensory Shift ($A_t \rightarrow W_{t+1} \rightarrow O_{t+1}$)